Storm Chasers

Severe weather like thunderstorms, tornadoes, hurricanes, blizzards, and floods happens when the air around Earth changes. You will learn how these storms form, how scientists predict them, how to stay safe, and how storms change the land, plants, and animals.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 9	Five Kinds of Severe Weather
Lesson 2	Fri Nov 13	Thunderstorms: Lightning, Hail, and Wind
Lesson 3	Fri Jan 8	Tornadoes and Hurricanes
Lesson 4	Fri Feb 5	Blizzards, Floods, and a Changing Land
Lesson 5	Fri Mar 5	Mock Tournament
Lab I1	Before Lesson 3	Weather Station Week: Tools, Data, and Maps (home or an open Friday)
Lab I2	Before Lesson 5	Watch, Warning, and Ready: Our Family Storm Plan (home or an open Friday)

At the tournament

Partners rotate through timed stations for up to 50 minutes (no going back) answering questions about severe weather from pictures, maps, data, and scenarios; they bring only pencils, so everything must be memorized.

Keep every handout for this event behind this divider, in order. Bring this binder to every Field B Fridays meeting.

Storm Chasers: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can identify thunderstorms, tornadoes, hurricanes, blizzards, and floods from a picture or description.	L1 L5	☐	☐
I can describe the main features of each type of severe weather and compare them (size, how long it lasts, where it forms).	L1 L3 L4 L5	☐	☐
I can explain that severe weather comes from changes in the atmosphere (warm and cold air, moisture, air pressure) and name conditions that lead to storms.	L1 L2 L3 I1	☐	☐
I can explain how a thunderstorm forms and name its hazards: lightning, thunder, hail, strong winds, and heavy rain.	L2 L5	☐	☐
I can describe a tornado and use the EF scale to rate its damage.	L3 L5	☐	☐
I can label a hurricane (eye, eyewall, rain bands), explain how it forms over warm ocean water, and describe storm surge.	L3 L5	☐	☐
I can describe blizzards, floods, and flash floods and explain why 'Turn Around, Don't Drown' matters.	L4 L5	☐	☐
I can read a simple weather map: fronts, H and L, radar colors, and a forecast.	L5 I1	☐	☐
I can explain the difference between a weather watch and a weather warning.	L5 I2	☐	☐
I can tell what to do before, during, and after each kind of severe weather to stay safe.	L2 L3 L4 L5 I2	☐	☐
I can explain how severe weather affects people, homes, schools, and communities, and how communities prepare and respond.	L4 I2	☐	☐
I can describe how flooding, wind, lightning, and heavy rain or snow affect plants, animals, and habitats.	L4 L5	☐	☐
I can explain how severe weather causes erosion and changes landforms like beaches, riverbanks, and hills.	L4 L5	☐	☐
I can name tools meteorologists use (thermometer, barometer, anemometer, wind vane, rain gauge, radar, satellites, weather balloons) and what each measures.	L5 I1	☐	☐
I can use a real storm, like Hurricane Florence in NC, to explain what severe weather does to people and the environment.	L3 L4 L5	☐	☐
I can use data, maps, pictures, and scenarios to solve severe weather problems.	L2 L5 I1 I2	☐	☐

Five Kinds of Severe Weather

TODAY I CAN

- I can identify thunderstorms, tornadoes, hurricanes, blizzards, and floods from a picture or description.
- I can describe the main features of each type of severe weather and compare them (size, how long it lasts, where it forms).
- I can explain that severe weather comes from changes in the atmosphere (warm and cold air, moisture, air pressure) and name conditions that lead to storms.

Storm Chasers is about severe weather: dangerous weather that can hurt people and change the land. At the tournament you cannot use notes, so you will learn these facts by heart.

Today's steps: run it yourselves (22 minutes)

You need: Storm photo cards (2 sets of 5), Clue cards (2 sets of 15), Blank index cards (30), Snack-size zip bags (2), Lesson 1 answer card (Event Captain). Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading the intro out loud. Tell your partner which storm you starred in the Do Now and why.
2. Both 7 min. Storm Sort: fold this page so the table is hidden. Lay the 5 photo cards in a row. Partner A reads a clue card and Partner B places it under a storm, 3 clues each. Switch jobs halfway.
3. Both 3 min. Unfold the page. Check your sort with 'The 5 kinds of severe weather' table and fix it. Find 2 storms that can happen at the same time.
4. Partner A 4 min. Read this out loud: 'The atmosphere is the layer of air around Earth. When warm, wet air rises into cold air, clouds grow and storms can form.' Then both fill in the 'Fill in' box with the word bank.
5. Both 2 min. Answer the 'Quick check' questions without looking at the table.
6. Both 3 min. Make 5 flashcards: storm name on the front, 3 clues on the back. Put them in a zip bag in your binder pocket.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Check your Fill in and Quick check answers. Fix any you missed in a different color.

The 5 kinds of severe weather

Storm	Clue 1	Clue 2	Clue 3
Thunderstorm	Has lightning and thunder	Tall cloud called a cumulonimbus	Most common in summer afternoons
Tornado	Spinning column of air touching the ground	Forms from a thunderstorm	Usually lasts only minutes
Hurricane	Forms over warm ocean water	Has a calm eye in the middle	Winds of 74 mph or more
Blizzard	Snow with winds of 35 mph or more	So hard to see you cannot see 1/4 mile	Lasts 3 hours or more
Flood	Water covers land that is usually dry	Flash floods happen in minutes	Caused by heavy rain or ocean water pushed ashore

Fill in

All severe weather comes from changes in the _____ (the layer of air around Earth). Many storms start when _____, wet air rises into _____ air above it.

Word bank

atmosphere **warm** **cold**

Quick check

1. A storm with an eye that forms over warm ocean water is a...

◯ Tornado ◯ Hurricane ◯ Blizzard

2. A spinning column of air that touches the ground is a...

◯ Tornado ◯ Flood ◯ Thunderstorm

3. Strong wind, blowing snow, and you cannot see far. This is a...

◯ Hurricane ◯ Blizzard ◯ Flood

4. Water covers a road that is usually dry. This is a...

◯ Blizzard ◯ Tornado ◯ Flood

At home before next time: Study your 5 flashcards: read the front, say the 3 clues, then flip to check. Do this 3 times this week. Then interview an adult about a big storm they remember (hurricane, snowstorm, flood, or tornado) using the Storm Story Interview page in your Storm Watcher Weather Journal (p. 13). That earns the Storm Storyteller badge. Bring your journal and flashcards to Lesson 2 on Nov 13. Bonus: start the Cloud Spotter page (p. 5).

Thunderstorms: Lightning, Hail, and Wind

TODAY I CAN

- I can explain that severe weather comes from changes in the atmosphere (warm and cold air, moisture, air pressure) and name conditions that lead to storms.
- I can explain how a thunderstorm forms and name its hazards: lightning, thunder, hail, strong winds, and heavy rain.
- I can tell what to do before, during, and after each kind of severe weather to stay safe.
- I can use data, maps, pictures, and scenarios to solve severe weather problems.

Thunderstorms are the most common severe weather. Every thunderstorm has lightning. Learn these facts by heart.

Today's steps: run it yourselves (22 minutes)

You need: Clear container, Blue ice cubes (in the small cooler), Cup, red food coloring, and dropper, Flashlight, Stage cards (2 sets of 3), Lesson 2 answer card (Event Captain). Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Read 'How far?' in the 'Lightning safety' box. Check your Do Now math with your partner. Is that close enough to be dangerous?
2. Both 6 min. Fill the container with room-temperature water, and ask the adult in the room for a cup of hot tap water with red coloring. Partner A drops a blue ice cube in one end while Partner B squirts red water at the bottom of the other end. Which color rises? Read 'Updraft' and 'Cumulonimbus' to see why.
3. Both 6 min. Lightning game: Partner A flashes the flashlight, counts silently, then knocks on the table. Partner B counts from flash to knock, divides by 5, and writes the miles on the whiteboard. Try 5, 15, and 25 seconds. Switch jobs.
4. Both 4 min. Put the 3 stage cards in order and read the backs. Check with '3 stages of a thunderstorm'. Read the 'Hail' row: how does hail get so big?
5. Both 3 min. Make 4 new flashcards: lightning, thunder, hail, updraft. Take an Independent Lab I1 page. The Check yourself questions are for home.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Check your Do Now math, your lightning game miles, and your stage order. Fix any you missed in a different color.

Thunderstorm facts

Ingredients	Moisture (water vapor), warm air that rises, and something to lift it (like a cold front or the hot sun)
Cumulonimbus	The tall, dark thunderstorm cloud. The top can spread out like an anvil
Updraft	Warm air rushing up into the storm
Downdraft	Cool air and rain rushing down. Can cause damaging straight-line winds
Lightning	A giant spark of electricity in a storm
Thunder	The sound lightning makes when it heats the air so fast the air expands with a boom
Hail	Balls of ice. Updrafts carry them up and down, adding layers of ice
Severe thunderstorm	Hail 1 inch or bigger, winds 58 mph or more, or a tornado

3 stages of a thunderstorm

1. Cumulus (developing): warm air rises and a cloud grows taller. Updrafts only.
2. Mature: heavy rain, lightning, hail, and strong winds. Updrafts and downdrafts. Most dangerous stage.
3. Dissipating: downdrafts take over, rain gets lighter, and the storm dies out.

Lightning safety

Rule	When thunder roars, go indoors
Where	Inside a building or a hard-top car. Not under a tree
After	Wait 30 minutes after the last thunder before going back outside

How far?	Count seconds from flash to boom, then divide by 5 = miles away

Check yourself (no peeking!)

1. How does a thunderstorm cloud form?

2. Name the 3 stages in order.

3. Which stage is the most dangerous?

4. What causes thunder?

5. You count 20 seconds from flash to boom. How far away? What should you do?

6. Where is a safe place during lightning?

7. What makes a thunderstorm 'severe'?

At home before next time: Do Independent Lab I1 (Weather Station Week) before Lesson 3 on Jan 8. Record your 7 days in the Weather Station Log in your Weather Journal (p. 6) for the Weather Watcher badge. Also cover the top of this page and answer the Check yourself questions from memory, check them, and practice all 9 flashcards with a family member.

Tornadoes and Hurricanes

TODAY I CAN

- I can describe the main features of each type of severe weather and compare them (size, how long it lasts, where it forms).
- I can explain that severe weather comes from changes in the atmosphere (warm and cold air, moisture, air pressure) and name conditions that lead to storms.
- I can describe a tornado and use the EF scale to rate its damage.
- I can label a hurricane (eye, eyewall, rain bands), explain how it forms over warm ocean water, and describe storm surge.
- I can tell what to do before, during, and after each kind of severe weather to stay safe.
- I can use a real storm, like Hurricane Florence in NC, to explain what severe weather does to people and the environment.

Tornadoes and hurricanes both have spinning winds, but they are very different in size and in how they form.

Today's steps: run it yourselves (27 minutes)

You need: Tornado bottle and towel, Tornado damage cards (4, numbered), Hurricane formation cards (2 envelopes), Lesson 3 answer card (Event Captain). Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Partner A quizzes Partner B on 5 flashcards from Lessons 1 and 2 and Lab I1. Then switch.
2. Both 3 min. Each partner swirls the tornado bottle once over the towel. What shape do you see? Take turns reading the 'Tornado facts' box.
3. Both 4 min. Rate the 4 damage photos with the EF scale in the Tornado facts box (EF0 is light damage, EF5 is houses swept away). Write the EF number on a sticky note on each photo.
4. Both 7 min. Put the hurricane cards in order. Then read the 'Hurricane facts' box and draw a hurricane from above in the box. Label the eye, eyewall, and rain bands. Why do hurricanes get weaker over land?
5. Both 5 min. Fill in the 'Compare' table for tornadoes and hurricanes. Use both facts boxes.
6. Both 4 min. Read the 2 'Safety' rows out loud. Make 4 flashcards: EF scale, eye, storm surge, tornado safety. The Florence questions are for home.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Check your damage ratings, hurricane labels, and Compare table. Fix any you missed in a different color.

Tornado facts

Tornado	A spinning column of air that reaches from a thunderstorm cloud to the ground
Funnel cloud	A spinning cloud that has not touched the ground yet
Supercell	A strong, rotating thunderstorm that makes most strong tornadoes
EF scale	Rates tornadoes by damage: EF0 (light damage, 65-85 mph) up to EF5 (houses swept away, over 200 mph)
Safety	Go to the lowest floor, in a small inside room with no windows. Get down and cover your head

Hurricane facts

How it forms	Over warm ocean water (about 80 degrees F). Warm, wet air rises and makes thunderstorms that start to spin
Growing up	Tropical depression (up to 38 mph), tropical storm (39-73 mph), hurricane (74 mph or more)
Eye	Calm, clear center
Eyewall	Ring of storms around the eye with the strongest winds
Rain bands	Spiral arms of storms and heavy rain

Categories	Category 1 (74 mph) to Category 5 (157 mph or more)
Storm surge	Ocean water pushed onto land by the wind. The most dangerous part near the coast
Season	June 1 to November 30 in the Atlantic
Safety	Have a kit, leave if officials say to evacuate, stay away from windows

Draw a hurricane from above and label: eye, eyewall, rain bands

Compare

	Tornado	Hurricane
Size		
How long it lasts		
Where it forms		
Scale used		

Real storm: Hurricane Florence (2018)

Florence came ashore near Wrightsville Beach, NC, in September 2018 as a Category 1 hurricane. Its winds were not the worst part. It moved very slowly, so it dropped rain on the same places for days. Some towns got more than 30 inches of rain. Rivers flooded homes, roads, and farms. Storm surge pushed ocean water into coastal towns and washed away beach sand.

Think about Florence

1. What caused most of the damage from Florence?

2. Name 2 ways Florence changed people's lives or the land.

3. Why did a 'weak' Category 1 storm cause so much flooding?

At home before next time: Answer the Florence questions. Then practice your flashcards (you should have about 18 now) until you can say every one without looking. Challenge: explain to a family member the difference between the eye and the eyewall, and where to go in your house during a tornado warning. Weather Journal: use the Beaufort scale (p. 8) on 5 days for the Wind Watcher badge, and finish spotting all 5 cloud types (p. 5) for Cloud Spotter.

Blizzards, Floods, and a Changing Land

TODAY I CAN

- I can describe the main features of each type of severe weather and compare them (size, how long it lasts, where it forms).
- I can describe blizzards, floods, and flash floods and explain why 'Turn Around, Don't Drown' matters.
- I can tell what to do before, during, and after each kind of severe weather to stay safe.
- I can explain how severe weather affects people, homes, schools, and communities, and how communities prepare and respond.
- I can describe how flooding, wind, lightning, and heavy rain or snow affect plants, animals, and habitats.
- I can explain how severe weather causes erosion and changes landforms like beaches, riverbanks, and hills.
- I can use a real storm, like Hurricane Florence in NC, to explain what severe weather does to people and the environment.

Blizzards and floods are severe weather too. Heavy rain, wind, and waves can also move soil and sand and change the shape of the land.

Today's steps: run it yourselves (27 minutes)

You need: Pan A (bare) and Pan B (grass), Rain cups (2) and pitcher, Thick books (2), Old towels or trays, Lesson 4 answer card (Event Captain). Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Quiz each other on 5 flashcards each. Then ask one Florence question from the Lesson 3 page.
2. Both 4 min. Take turns reading the 'Blizzards and floods' box out loud. Tell your partner why driving through flood water is dangerous (see 'Flood safety').
3. Both 3 min. Set up: put towels under both pans. Put a book under the hill end of each pan to tilt it. Fill the pitcher with water.
4. Both 9 min. Partner A slowly pours a full rain cup over the top of the hill in Pan A for 30 seconds while Partner B watches. Do the same for Pan B. Then pour half the pitcher fast on each pan (a flash flood). Fill in the 'Erosion experiment' table.
5. Both 4 min. Read the 'How severe weather changes the environment' table. Then answer the 'Check yourself' questions.
6. Both 3 min. Make 3 flashcards: blizzard, flash flood, erosion. Take an Independent Lab I2 page.

Check your work: Ask your Event Captain for the Lesson 4 answer card. Check your experiment table and Check yourself answers. Fix any you missed in a different color.

Blizzards and floods

Blizzard	Winds of 35 mph or more and blowing snow, so you cannot see 1/4 mile, for 3 hours or more
Blizzard safety	Stay inside, dress in layers, keep extra blankets, food, and flashlights
Flood	Water covers land that is usually dry, from heavy rain, melting snow, or storm surge
Flash flood	A flood that happens very fast, in minutes to a few hours
Flood safety	Turn Around, Don't Drown! Just 6 inches of moving water can knock you down; 12 inches can carry away a car
Erosion	When water, wind, or ice carries soil and rock away
Deposition	When the moving soil and sand are dropped in a new place

Erosion experiment

	Pan A: bare soil	Pan B: with grass
Where did the soil go?		
How muddy was the water? (1-5)		
Did gullies form?		

How severe weather changes the environment

Cause	Effects
Flooding	Washes away soil, carves riverbanks, drowns burrows; leaves new rich soil on floodplains
Strong wind	Knocks down trees and nests; fallen trees become new homes for insects and animals
Lightning	Can start wildfires; some pine forests need fire to stay healthy
Heavy rain or snow	Erodes hills, causes mudslides, breaks branches
Storm surge	Washes away beaches and dunes; salt water kills marsh plants

Check yourself

1. What 3 things make a storm a blizzard?

2. What is a flash flood?

3. What does 'Turn Around, Don't Drown' mean?

4. Why did the grass pan lose less soil?

5. Name one landform that heavy rain or storm surge can change.

6. Name one way severe weather affects an animal's habitat.

At home before next time: Do Independent Lab I2 (Watch, Warning, and Ready) with your family before Lesson 5 on Mar 5, and fill in Our safe places on the Watch or Warning? page of your Weather Journal (p. 15) for the Safety Captain badge. Then have someone quiz you on ALL your flashcards (about 21) and put a check on the ones you know cold. Optional: on a walk, look for one spot where rain has washed soil away and fill in the Erosion Detective page (p. 16). Next time is our mock tournament, with no notes.

Mock Tournament

TODAY I CAN

- Everything on my can-do checklist (the page right after the divider). Today is the time to show it.

This is a practice tournament. No notes. Work with your partner and answer every question, even if you have to guess.

Today's steps: run it yourselves (27 minutes)

You need: Station signs (5), Station 1 envelope (5 storm photos), Station 2 envelope (weather map), Station 3 envelope (3 damage photos), Lesson 5 answer card (Event Captain). Set the timer for each step.

1. Event Captain 2 min. Collect all binders and flashcards into the bin (no notes) and put the station signs and envelopes at 5 spots. The Event Judge reads the intro out loud: 4 minutes per station, no going back. Set the timer for 4 minutes.
2. Both 8 min. Stations 1 and 2. The Event Judge calls 'One minute!' at 3 minutes and 'Rotate!' at 4, then resets the timer.
3. Both 12 min. Stations 3, 4, and 5. Same timing. Answer every question, even with a guess. If you finish early, check your work.
4. Event Captain 3 min. Hand out the answer card. Each pair scores its own sheet.
5. Both 2 min. Circle the 2 topics you missed most. Write them on the 'Study before the tournament' line.

Check your work: Use the Lesson 5 answer card from your Event Captain to score your sheet, 1 point per right answer. Fix misses in a different color, then make flashcards for them at home.

Station 1: Name that storm

Photo #	Type of severe weather
1	
2	
3	
4	
5	

1. Give 2 differences between a hurricane and a tornado.

Station 2: Weather map

1. What is the blue line with triangles?

2. Where are storms most likely on this map? Why?

3. What weather does the H usually bring?

Station 3: Storm damage detective

1. For each photo, name the cause (wind, flood, storm surge, etc.).

2. What should you do if you see a flooded road?

Station 4: Safety and science

1. The radio says a tornado was seen heading toward your town. Watch or warning? What do you do?

2. You count 15 seconds from lightning to thunder. How far away? What do you do?

3. Which tool measures wind speed?

Station 5: Storms and the land

1. Why does a hill with grass lose less soil in a storm than a bare hill?

2. Name one way a hurricane can change an animal's habitat.

3. What NC storm did we study, and what was its biggest effect?

4. Study before the tournament: our 2 weakest topics

At home before next time: Make 3 new flashcards for your 2 weakest topics. Practice all your flashcards 3 times a week until the tournament. Teach one topic to a family member without looking. If you can teach it, you know it. Weather Journal: copy a real weather map with a front, H, and L from the news or an app (p. 14) for the Map Reader badge.

Lab I1: Weather Station Week: Tools, Data, and Maps

THIS LAB HELPS ME

- I can explain that severe weather comes from changes in the atmosphere (warm and cold air, moisture, air pressure) and name conditions that lead to storms.
- I can read a simple weather map: fronts, H and L, radar colors, and a forecast.
- I can name tools meteorologists use (thermometer, barometer, anemometer, wind vane, rain gauge, radar, satellites, weather balloons) and what each measures.
- I can use data, maps, pictures, and scenarios to solve severe weather problems.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Meteorologists are scientists who study weather. They measure the air every day and put their data on maps. This week you are the meteorologist.

Steps (28 minutes)

You need: Clear cup or jar and a ruler, Blue, red, and green colored pencils, 5 index cards. Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Read 'Meteorologist tools' out loud. Then cover the right side and say what each tool measures.
2. Me 3 min. Set up your rain gauge: put the cup or jar outside in an open spot, away from roofs and trees. Put a rock next to it so it does not tip. (On an open Friday, take the cup home.)
3. Me 8 min. For 4 days, take 2 minutes each day to fill in one row of the 'Weather log'. Measure the rain in the cup with a ruler, then empty it. Look at a flag or tree to see where the wind comes from.
4. Me 8 min. Set the timer for 8 minutes. Read 'Weather map symbols'. Follow 'Draw a weather map' step by step in the box. Then answer 'Read your map'.
5. Me 5 min. Set the timer for 5 minutes. Answer 'Check yourself' without looking. Then make 5 flashcards: barometer, anemometer, wind vane, cold front, L.

Check your work: Check 'Read your map' and 'Check yourself' with the 'Meteorologist tools' and 'Weather map symbols' boxes. On an open Friday or at Lesson 3, ask your Event Captain for the Lab I1 answer card and fix any you missed in a different color.

Meteorologist tools

Thermometer	Temperature
Barometer	Air pressure (falling pressure often means a storm is coming)
Anemometer	Wind speed (spinning cups)
Wind vane	Wind direction
Rain gauge	How much rain fell
Doppler radar	Where rain, hail, and snow are, and if a storm is rotating
Weather satellite	Pictures of clouds and storms from space
Weather balloon	Weather high up in the atmosphere, launched twice a day

Weather log

Day	Temperature (or warm, cool, cold)	Rain in cup (cm)	Wind comes from	Sky and clouds
1				
2				
3				
4				

What my data shows

1. Which day was the stormiest? What clue in your log told you?

Weather map symbols

Cold front	Blue line with triangles. Cold air pushes in; can bring thunderstorms, then cooler air

Warm front	Red line with half circles. Warm air moves in; clouds and steady rain
Stationary front	Red and blue line that alternates triangles and half circles. Not moving; days of rain
H	High pressure: usually clear, calm weather
L	Low pressure: usually clouds, rain, and storms
Radar colors	Green = light rain, yellow = moderate, red = heavy, purple or pink = very heavy or hail

Draw a weather map

1. Draw a big outline of land that fills most of the box.
2. Near the top left, write a big blue H. Put a small letter C next to it.
3. In the middle, write a big red L. Put a small letter D next to it.
4. Cold front: from the L, draw a blue line going down and to the left. Add blue triangles on its right side. Label it A.
5. Warm front: from the L, draw a red line going to the right. Add red half circles on its top side. Label it B.
6. Radar: color a green patch along the cold front. Inside it, color a small red patch. Label the red patch E.

My weather map

Read your map

1. Name A, B, C, D, and E.

2. Where on your map are storms most likely? Why?

3. What weather is usually under the H?

4. What happens after a cold front passes?

Check yourself

1. Which tool measures wind speed? Which shows wind direction?

2. Falling numbers on a barometer often mean what?

3. Which tool can see if a storm is rotating?

Bring to the next meeting: Bring this page and your 5 new flashcards to Lesson 3 on Jan 8. You should have about 14 flashcards now.

Lab I2: Watch, Warning, and Ready: Our Family Storm Plan

THIS LAB HELPS ME

- I can explain the difference between a weather watch and a weather warning.
- I can tell what to do before, during, and after each kind of severe weather to stay safe.
- I can explain how severe weather affects people, homes, schools, and communities, and how communities prepare and respond.
- I can use data, maps, pictures, and scenarios to solve severe weather problems.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Forecasters send out watches and warnings so people have time to get ready. In this lab you practice telling them apart and make a real storm plan with your family.

Steps (27 minutes)

You need: This page, A pencil. Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read 'Watch or warning?' out loud. Write the rule in your own words at the bottom of the box.
2. Me 7 min. Set the timer for 7 minutes. For each alert in the 'Watch or Warning' table, write Watch or Warning and what you would do.
3. Me 8 min. With a family member, go through the 'Emergency kit check'. Check each thing your home has. Circle what is missing. On an open Friday, star what you think you have and finish at home.
4. Me 5 min. Fill in 'Our family plan' with your family: where to go for each storm, a meeting spot, and an adult to call.
5. Me 4 min. Set the timer for 4 minutes. Read 'Getting ready: families and communities'. Answer the 'Check yourself' questions.

Check your work: Check the 'Watch or Warning' table with the rule in the 'Watch or warning?' box. On an open Friday or at Lesson 5, ask your Event Captain for the Lab I2 answer card and fix any you missed in a different color.

Watch or warning?

Watch	Conditions are right for the storm. Be prepared and keep listening.
Warning	The storm is happening now or is about to. Take action now.
Tip	Words like 'possible' or 'could' mean watch. Words like 'seen', 'happening', or 'moving in now' mean warning.
My rule	_____

Watch or Warning

Alert	Watch or Warning?	What I would do
1. Conditions are right for tornadoes this afternoon.		
2. A tornado was seen on radar heading to your town.		
3. Flash flooding is happening on Main Street.		
4. Heavy snow and strong wind are possible tomorrow.		
5. Hurricane conditions are expected within 36 hours.		
6. Severe thunderstorms could form tonight.		
7. A severe thunderstorm with 60 mph winds is moving in now.		
8. Hurricane conditions are possible within 48 hours.		

Emergency kit check

- ☐ Water (1 gallon per person per day, for 3 days)
- ☐ Food that will not spoil, and a can opener
- ☐ Flashlight and extra batteries
- ☐ First aid kit and medicine
- ☐ Battery or crank radio (a NOAA Weather Radio is best)

- [ ] Phone charger
- [ ] Blankets and warm clothes
- [ ] Copies of important papers in a zip bag

Our family plan

If there is a…	We go to…
Tornado warning	
Thunderstorm (lightning)	
Flood warning	
Meeting spot if we are split up	
Adult to call	

Getting ready: families and communities

Emergency kit	Water, food that will not spoil, flashlight, batteries, first aid kit, medicine, weather radio
Family plan	Know your safe place, a meeting spot, and an adult to call
Alerts	Sirens, phone alerts, NOAA Weather Radio, TV and radio
Community	Shelters, school drills, clearing storm drains, sandbags, rescue and repair crews after the storm
After a storm	Stay away from downed power lines and flood water; wait for adults to say it is safe

Check yourself

1. Name 2 ways people get storm alerts.

2. Name 2 ways a community prepares for or responds to storms.

3. After a storm, what 2 things should you stay away from?

Bring to the next meeting: Show your family plan to everyone at home and put a copy on the fridge. Bring this page to Lesson 5 on Mar 5.